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Effective management of eclampsia in bitches – A veterinary emergency

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Abstract

Eclampsia, or puerperal tetany, is a serious and potentially fatal disorder of lactating bitches characterized by acute hypocalcaemia. A group of six lactating bitches of small breeds aged between 1.5 and 3 years and weighing 4–6 kg were presented to the Veterinary Clinical Complex, College of Veterinary Science, Rajendranagar, Hyderabad, with clinical signs, including markedly elevated body temperature (106°F to 109°F), lateral recumbency (33.3%), respiratory distress (50%), tachycardia (66.6%), excessive panting (33.3%), muscular tremors (100%), and jaw chattering (100%). Serum biochemical analysis revealed significantly reduced calcium levels having a mean value of 4.55mg/dl, confirming postpartum eclampsia. Diagnosis was established based on history, characteristic clinical signs, and laboratory findings. Treatment included slow intravenous administration of 10% calcium gluconate at 0.5 ml/kg body weight, along with Dexamethasone (0.04 mg/kg), Paracetamol (10 mg/kg) (The Latest in Pain Management - WSAVA2013 – VIN), and Dextrose normal saline (10 ml/kg) as supportive therapy. Following treatment, all bitches exhibited a rapid return to normal body temperature (100–101°F) and complete resolution of clinical symptoms. This case series highlights the importance of early detection and immediate therapeutic intervention in managing postpartum hypocalcaemia, especially in small-breed dogs, to prevent life-threatening complications and ensure successful recovery.

Keywords: Dog, bitches, post-partum eclampsia, hypocalcaemia, 10% calcium gluconate

Introduction

Eclampsia is a serious and potentially fatal condition caused by hypocalcaemia, most commonly seen in lactating bitches and rarely reported in cats. Unlike in cattle, cases of hypocalcaemia in small animals are less frequently documented (Rowland *et al.*, 1972) [5]. The disorder typically occurs one to three weeks after whelping, although it can sometimes occur in the later stages of pregnancy. Several predisposing factors increase the risk of eclampsia, including breeding at the first oestrus, small body size (as smaller breeds have higher calcium demands relative to their size), young age combined with large litter size, poor nutritional management, and stressful environmental conditions (Pathan *et al.*, 2011) [4]. The disease develops when calcium loss through milk exceeds the mother's ability to replenish blood calcium levels.

During the early lactation phase, increasing milk production and the calcium requirements of the growing puppies intensify this imbalance. The postpartum period up to 40 days is particularly critical. Interestingly, the puppies generally remain unaffected, as milk calcium levels are usually sufficient to meet their needs (Pathan *et al.*, 2011) [4]. In small breeds, calcium homeostasis mechanisms often fail to compensate quickly enough, resulting in hypocalcaemia which will be manifested as restlessness, increased body temperature, rapid heartbeat, muscle tremors, incoordination, and convulsions (Ettinger, 1983) [2].

Materials and Methods

A group of six lactating bitches of small breeds, aged between 1.5 and 3 years and weighing 4–6 kg were presented to the Veterinary Clinical Complex, College of Veterinary Science, Rajendranagar, Hyderabad. On clinical examination the animals exhibited pyrexia (106°-

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109°F), lateral recumbency (33.3%), respiratory distress (50%), tachycardia (66.6%), panting (33.3%), muscle tremors (100%) and jaw champing (100%). In this group study, six lactating small-breed bitches that had whelped approximately 25–30 days earlier were presented, each having delivered between four to six healthy puppies. At the time of hospitalisation, in sterile vials with and without EDTA about 4ml of blood samples were collected from the cephalic vein using a 24G needle for Complete Blood Count (CBC) and estimation of serum calcium and phosphorus levels. Laboratory evaluation indicated marked hypocalcaemia in all six animals, with serum calcium concentrations ranging from 3.8 to 5.3 mg/dl (reference range: 9.1–11.7 mg/dl). Serum phosphorus values were within or slightly above normal limits (4.8–5.3 mg/dl; reference range: 2.9–5.3 mg/dl).

Haematological parameters for all bitches remained within normal physiological limits. All six bitches were having markedly elevated temperature (106–109°F) at admission, initial management aimed at temperature control. Cold packs were applied (Fig. 1), and cold-water enemas were administered every 10 minutes. Dextrose Normal Saline (5%) was given intravenously at 10 ml/kg body weight, and Paracetamol was administered intravenously at 10 mg/kg body weight (The Latest in Pain Management - WSAVA2013 – VIN). Vital parameters (temperature and heart rate) were recorded at 5-minute intervals. Within the first hour of treatment, body temperature in all animals decreased to approximately 102.5–103°F. After stabilization of hyperthermia, 10% calcium gluconate was administered slowly intravenously at 0.5 ml/kg body weight, with continuous monitoring of cardiac rhythm by auscultation. Within 30 minutes of calcium therapy, body temperature further stabilized to around 100.8–101°F (Fig.2), and clinical signs such as muscular tremors, seizures, and jaw champing resolved completely in all five bitches.



Fig 1: Application of cold packs



Fig 2: Temperature during and after treatment.

Results and Discussion

Based on history, clinical examination, and diagnostic evaluation, the condition was diagnosed as post-partum eclampsia. Blood analysis revealed decreased blood calcium concentrations having a mean value of 4.55 mg/dl, while other haematological parameters were within normal limits.

After initiating appropriate treatment, five out of six animals showed complete recovery, and one was succumbed to death. To prevent recurrence, oral calcium along with vitamin and mineral supplementations were recommended for two weeks. Owners were advised to shift the puppies to milk replacer and weaning of all the puppies as soon as possible. The neurological responses and overall activities of the animals returned to normal. During the final evaluation, the animals were found to be in good health and was subsequently discharged, indicating a positive outcome and effective therapeutic management.

Table 1: Mean $\pm$ SE values of serum calcium and phosphorus in hypocalcemic lactating bitches (n=6)

Parameter	Affected bitches (n = 6)	Reference values*
Calcium (mg/dL)	4.55 $\pm$ 0.23	9.1 – 11.7
Phosphorus (mg/dL)	4.87 $\pm$ 0.11	2.9 – 5.3

*The reference values (normal ranges) are taken from the Merck Veterinary Manual.

Table 2: Mean $\pm$ SE values of normal hematological parameters in lactating bitches (n=6)

Parameter	Normal lactating bitches (n=6)	Reference values*
Hemoglobin (g/dL)	14.2 $\pm$ 0.35	9.3–15.3
Packed Cell Volume (%)	40.8 $\pm$ 0.90	28–49
Total Erythrocyte Count ($\times 10^6/\mu\text{L}$)	6.20 $\pm$ 0.18	4.6–10
Total Leukocyte Count ($\times 10^3/\mu\text{L}$)	10.7 $\pm$ 0.40	5.5–19.5
Neutrophils (%)	68.5 $\pm$ 1.20	35–85
Lymphocytes (%)	23.4 $\pm$ 0.85	12–45
Monocytes (%)	4.1 $\pm$ 0.20	2–9
Eosinophils (%)	2.8 $\pm$ 0.15	0

*The reference values (normal ranges) are taken from the Merck Veterinary Manual.

Eclampsia, or puerperal tetany, is a metabolic disorder most often seen in small-breed bitches during the first four weeks after whelping, when calcium requirements peak due to intense milk production. It occurs when the body's calcium-regulating mechanisms fail to meet the increased lactational demand, leading to severe hypocalcaemia. The condition is mainly associated with calcium depletion during milk production, worsened by nutritional deficiencies, inappropriate supplementation, or excessive calcium mobilization from bone. Homemade diets high in meat and low in calcium are a major predisposing factor, as meat has a low calcium-to-phosphorus ratio—sometimes as low as 1:15, compared with the ideal 1.2:1—which reduces calcium bioavailability (Simpson *et al.*, 1993) [6]. Low protein intake or excessive protein loss may also contribute to reduced calcium levels (Mellanby *et al.*, 2005) [3], while parathyroid disorders are rarely involved (Pathan *et al.*, 2011) [4].



Fig 3: Image of dog one-week post-recovery.

Clinically, eclampsia usually appears 10–30 days postpartum, presenting with tremors, muscle rigidity, restlessness, and tetany. Prompt treatment is vital for recovery. The intravenous administration of 10% calcium gluconate (0.5–1.5 mL/kg over 15–30 minutes) remains the standard therapy, with close monitoring for bradycardia or arrhythmias. This is followed by dexamethasone (0.04 mg/kg IM), oral calcium supplementation, and dietary management to prevent relapse. In this case, the bitches showed marked improvement within 15 minutes of intravenous therapy—muscle tremors subsided, rigidity decreased, and breathing normalized. Continued oral supplementation and nutritional correction led to complete recovery. Preventive measures such as providing a balanced diet and allowing the dam adequate rest from nursing are essential to avoid recurrence.

Conclusion

The present group study describes a condition of post-partum eclampsia in 1.5 to 3 years-old lactating bitches, caused by hypocalcaemia and was confirmed by clinical history and laboratory evaluation and successfully managed by intravenous calcium therapy. Adjusting the diet to maintain the correct calcium-to-phosphorus ratio, or providing high-quality premium food based on the animal's is the main preventive measure. In cases of eclampsia, administering a 10% calcium gluconate solution is a simple and effective treatment approach. Additionally, early weaning of the puppies plays a crucial supportive role in the overall management of eclampsia.

Conflict of Interest: Not available

Financial Support: Not available

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